

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 32

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <p><i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• Organic advantages: often waste products, locally available and cheap; add humus improving structure and water retention; increase soil biota; slow release reduces leaching (<b>up to 2</b>)</li> <li>• Organic disadvantages: nutrient composition cannot be controlled; released slowly so needs long-term planning; bulky with high water content so expensive to transport; cannot be applied to a growing crop; odour / pathogens (<b>up to 2</b>)</li> <li>• Inorganic advantages: composition controlled to match crop needs; nutrients released rapidly; concentrated and easy to apply (<b>up to 2</b>)</li> <li>• Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process); no organic matter so humus and biota decline; some toxic to worms; high solubility so leached causing eutrophication; raw materials (phosphate rock) finite (<b>up to 2</b>)</li> </ul> | 6     |
| 01.2     | <p><i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• Crops grown in a circulating nutrient solution (inert medium) in greenhouses; nutrient concentrations monitored and adjusted so nutrients are never limiting; abiotic factors controlled (<b>up to 2</b>)</li> <li>• Roots are small (all in contact with nutrients) so more growth goes into the harvested crop; no soil-borne pathogens or weeds; crop stays fresh with roots (<b>up to 2</b>)</li> <li>• Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping); high level of technical knowledge; only economic for high-value crops (<b>up to 2</b>)</li> </ul>                                                                                                                                                                                                                                                                                                       | 4     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use <b>(2)</b></li> <li>• Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught <b>(2)</b></li> <li>• Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught <b>(2)</b></li> </ul>                                                                                                                      | 6     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch <b>(1)</b></li> <li>• Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more <b>(1)</b></li> <li>• Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>03.1</b> | <p><i>Up to 5 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates <b>(up to 2)</b></li> <li>• B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches <b>(up to 2)</b></li> <li>• C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil <b>(up to 2)</b></li> <li>• Habitat diversity / structural variety increases the range of niches and hence species <b>(up to 2)</b></li> </ul> | 5     |
| <b>03.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks <b>(1)</b></li> <li>• Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) <b>(1)</b></li> <li>• Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer <b>(1)</b></li> <li>• Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership <b>(1)</b></li> </ul>                                                                                                                                                                 | 3     |
| <b>03.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• No species is completely dominant and food webs have many alternative links <b>(1)</b></li> <li>• So a change in the population of one species has a relatively small impact on the whole community <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| <b>04.1</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people <b>(1)</b></li> <li>• Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| <b>04.2</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Total impact = population size x the individual (per capita) impact of each person <b>(1)</b></li> <li>• Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| <b>04.3</b> | <i>Any 1 from the points below - 1 mark each, maximum 1.</i> <ul style="list-style-type: none"> <li>• Slash and burn agriculture <b>(1)</b></li> <li>• Fishing <b>(1)</b></li> <li>• Discharge of sewage / biodegradable pollutants <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| <b>05.1</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates <b>(1)</b></li> <li>• Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials <b>(1)</b></li> <li>• Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult <b>(1)</b></li> <li>• Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable <b>(1)</b></li> </ul> | 4     |
| <b>05.2</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life <b>(1)</b></li> <li>• They leach from landfill or are released by incineration; persistent, bioaccumulating <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| <b>05.3</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) <b>(1)</b></li> <li>• Easy identification of materials (coding) and separation of components so they can be recycled <b>(1)</b></li> <li>• Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) <b>(1)</b></li> <li>• Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle) <b>(1)</b></li> </ul>                                                                                                                                                                             | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
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| <b>06.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve <b>(1)</b></li> <li>• Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed <b>(1)</b></li> <li>• Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3     |
| <b>06.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Auger collects a core of known cross-sectional area and depth <b>(1)</b></li> <li>• So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| <b>06.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Minerals such as calcium carbonate break down releasing carbon dioxide <b>(1)</b></li> <li>• So mass is lost that is not organic matter <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
| <b>06.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter <b>(1)</b></li> <li>• Standardise bulb brightness / colour, distance and container <b>(1)</b></li> <li>• A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>07.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) <b>(1)</b></li> <li>• SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) <b>(1)</b></li> <li>• Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally <b>(1)</b></li> <li>• MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&amp;CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities <b>(1)</b></li> </ul> | 4     |
| <b>07.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) <b>(1)</b></li> <li>• Management agreements between owners and statutory bodies to maintain suitable conditions <b>(1)</b></li> <li>• But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
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| <b>07.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues <b>(1)</b></li> <li>• Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) <b>(1)</b></li> <li>• Benefits everyone so society pays through central funding; legislation cannot compel positive management <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <b>08.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• General physiological solvent: most biological reactions take place between substances dissolved in water <b>(1)</b></li> <li>• Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes <b>(1)</b></li> <li>• High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) <b>(1)</b></li> <li>• Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                           | 4     |
| <b>08.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) <b>(1)</b></li> <li>• Oxygen in the stratosphere absorbed UV, forming ozone (O<sub>3</sub>) in a dynamic equilibrium of O, O<sub>2</sub> and O<sub>3</sub> <b>(1)</b></li> <li>• The ozone layer reduced UV reaching the surface so organisms could survive out of the water <b>(1)</b></li> <li>• Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out <b>(1)</b></li> </ul>                                                                                                                                                             | 4     |
| <b>08.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Photosynthesis removed CO<sub>2</sub> (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels <b>(1)</b></li> <li>• This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <b>09.1</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>• Copying the structural / functional adaptations of living organisms in the design of manufactured items <b>(1)</b></li> <li>• Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use <b>(1)</b></li> <li>• Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) <b>(1)</b></li> <li>• Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) <b>(1)</b></li> <li>• Termite mound convection copied for natural ventilation of buildings; hollow strutted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners <b>(1)</b></li> </ul> | 4     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
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| <b>09.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology <b>(1)</b></li> <li>Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) <b>(1)</b></li> <li>Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours) <b>(1)</b></li> </ul> | 3     |
| <b>09.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals <b>(1)</b></li> <li>Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) <b>(1)</b></li> </ul>                                                                                                                                                                             | 2     |
| <b>09.4</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>Squid giant nerve cells - nerve function / heart disease, stroke; marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection <b>(1)</b></li> </ul>                                                                                                                                                                                                  | 1     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
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| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports</li> <li>• Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics</li> <li>• Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling</li> <li>• Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties)</li> <li>• Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control</li> <li>• Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes</li> <li>• Social: control of impacts on rural communities, Fairtrade, local food; consumer choices</li> <li>• Evaluation: yields vs sustainability, cost, need for policy support; judgement</li> </ul> | 9     |
| 10.2     | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 11.1     | <p><b>Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Marine productivity: light, nutrients, upwelling; where fisheries occur</li> <li>• Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts</li> <li>• Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons)</li> <li>• Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae</li> <li>• Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR</li> <li>• Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 338-351; AQA spec 3.5.2.</p> | 25    |



| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
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| 11.2     | <p><b>Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations</li> <li>• Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk</li> <li>• Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges)</li> <li>• Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding</li> <li>• Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision</li> <li>• Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.</p> | 25    |